
Reflection Without Reliable Accuracy Gains: Measuring Representation Drift and Trajectory Stabilization Under Self-Critique

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Abstract

Self-critique prompting is widely used to improve large language model reasoning, but it is still unclear whether critique changes internal computation in a structured way or only perturbs outputs. We study this question through a joint behavioral and mechanistic evaluation on math and commonsense reasoning tasks. We build an executable harness that compares five generation conditions (DIRECT, REWRITE, SELF-CRITIQUE-1, SELF-CRITIQUE-3, and EXTERNAL-CRITIQUE) while extracting residual-stream activations at multiple depths. We measure exact-match accuracy, cosine drift, centered kernel alignment (CKA), linear-probe separability, and multi-round trajectory stability.

On QWEN2.5-1.5B-INSTRUCT (50 examples), all revision conditions induce large internal movement at a deep layer (mean cosine drift 0.3439–0.4230), yet none yields statistically significant accuracy gains over DIRECT after Benjamini–Hochberg correction ($q = 0.802$ for all comparisons). Multi-round self-critique shows decreasing step drift ($0.3683 \rightarrow 0.3598 \rightarrow 0.3347$), consistent with partial trajectory convergence. Correctness separability improves only for SELF-CRITIQUE-3 at the deepest probed layer (AUC $0.818 \rightarrow 0.889$), while other conditions are mixed or degrade. A smaller QWEN2.5-0.5B-INSTRUCT run and a behavioral-only GPT-4.1 check show similar non-significant differences.

These results support a nuanced conclusion: reflection prompts can strongly reshape internal states without delivering reliable accuracy gains in low-sample settings. Mechanistic change and behavioral improvement should therefore be evaluated as separate axes.

1 Introduction

Self-critique has become a default tool for improving language model reasoning. In practice, many systems ask a model to draft, critique, and revise using methods such as REFLEXION and SELF-REFINE [Shinn et al., 2023, Madaan et al., 2023]. This workflow often improves outputs, but the mechanism is still unclear.

why does this matter? If reflection only re-samples text without structured internal change, then observed gains may be brittle and hard to generalize. If reflection instead reorganizes hidden representations, then mechanistic signals can guide more reliable self-improvement pipelines.

Recent work identifies latent reflection or self-awareness directions in activation space [Zhu et al., 2025, Yan et al., 2025, Tamoyan et al., 2025]. However, most analyses are static: they detect whether a representation exists, not how the representation moves over multiple critique rounds. Our main question is whether iterative self-critique drives convergence toward a more stable and correctness-separable reasoning manifold.

We address this gap with DRIFTCRITIQUE, an evaluation harness that measures behavior and internal geometry jointly. On sampled GSM8K and COMMONSENSEQA items, we compare DIRECT

generation to four revision variants and extract residual-stream vectors across depth. We then quantify (i) behavioral accuracy, (ii) activation drift and CKA similarity, (iii) probe-based separability of correct vs. incorrect answers, and (iv) round-by-round trajectory stability.

Our quantitative preview is mixed but informative. On QWEN2.5-1.5B-INSTRUCT with $n = 50$, all revision conditions produce substantial deep-layer drift (mean 0.3439–0.4230), but none improves accuracy significantly versus DIRECT after FDR correction (all $q = 0.802$). At the same time, SELF-CRITIQUE-3 improves deepest-layer probe AUC from 0.818 to 0.889, and round-wise step drift decreases from 0.3683 to 0.3347.

Our contributions are:

- We propose a reproducible protocol that couples behavioral evaluation with residual-stream geometry across critique conditions and rounds.
- We conduct a controlled comparison of rewrite, self-critique, and external-critique prompts on GSM8K and COMMONSENSEQA using open local models.
- We show that large representational drift can occur without statistically significant accuracy gains, motivating separate evaluation axes for mechanism and behavior.
- We provide evidence of partial convergence and condition-dependent separability improvements, clarifying when self-critique appears structurally meaningful.

The rest of this paper is organized as follows. Section 2 reviews relevant prompting and mechanistic work, section 3 details our setup, section 4 reports results, and section 5 discusses limitations and implications.

2 Related Work

Self-critique and iterative refinement. Prompt-based revision methods such as REFLEXION [Shinn et al., 2023] and SELF-REFINE [Madaan et al., 2023] show that multi-step critique can improve generation quality without parameter updates. QUIET-STAR extends this direction by encouraging latent reasoning traces during generation [Zelikman et al., 2024]. Our work is complementary: instead of proposing a new prompting algorithm, we test whether critique prompts induce measurable internal-state structure beyond output changes.

Representation-level analyses of reflection and self-awareness. Recent studies report identifiable reflection-related directions in hidden states and use representation engineering to steer this behavior [Zhu et al., 2025, Yan et al., 2025]. Parallel work on factual self-awareness shows that linear features in residual activations can predict correctness [Tamoyan et al., 2025]. Additional findings on self-generated text recognition also point to interpretable residual-stream signatures [Ackerman and Panickssery, 2024]. Building on this line, we focus on *dynamics*: we measure how representations drift and stabilize across critique rounds.

Evaluation and reliability benchmarks. Reasoning and truthfulness benchmarks such as GSM8K [Cobbe et al., 2021a], COMMONSENSEQA [Talmor et al., 2019], and TRUTHFULQA [Lin et al., 2021] provide task-level outcomes for checking whether internal changes track performance. Work on verifiers further motivates separating process quality from final-answer quality [Cobbe et al., 2021b]. Our experiments adopt this separation directly by reporting both behavioral metrics and mechanistic metrics under matched prompts.

Positioning. Unlike prior static probing studies, we analyze multi-condition, multi-round trajectory behavior with paired statistical tests. Unlike prompting-only studies, we quantify residual-stream drift, CKA similarity, and correctness separability together. This combination directly addresses whether self-critique corresponds to structured internal reorganization.

3 Methodology

3.1 Research Question and Experimental Design

We test whether self-critique induces structured internal changes rather than only output variation. For each question, we generate one DIRECT draft and up to four revisions: REWRITE control, SELF-

Table 1: Accuracy across prompting conditions. Best value per model block is in **bold**.

Model	DIRECT	REWRITE	SELF-CRITIQUE-1	SELF-CRITIQUE-3	EXTERNAL-CRITIQUE
QWEN2.5-1.5B-INSTRUCT ($n = 50$)	0.26	0.30	0.28	0.22	0.34
QWEN2.5-0.5B-INSTRUCT ($n = 24$)	0.042	0.167	0.083	0.042	0.25

CRITIQUE-1, SELF-CRITIQUE-3, and EXTERNAL-CRITIQUE. We then compare both answers and hidden representations between pre- and post-revision states.

3.2 Datasets and Splits

We use local copies of two reasoning datasets: GSM8K test split (1,319 items) and COMMONSENSEQA validation split (1,221 items). The main mechanistic run uses a deterministic subset of 25 GSM8K + 25 COMMONSENSEQA examples ($n = 50$). A size-comparison run uses 12 + 12 examples ($n = 24$). A behavioral-only API run uses 20 sampled items.

Data quality checks reported zero missing values, zero detected duplicates, and complete gold-label parse success for GSM8K (1319/1319). For COMMONSENSEQA, class counts are near balanced (A:239, B:255, C:241, D:251, E:235).

3.3 Models and Implementation

Local mechanistic runs use QWEN2.5-1.5B-INSTRUCT (primary) and QWEN2.5-0.5B-INSTRUCT (size comparison). Behavioral triangulation uses GPT-4.1 via API. We run deterministic decoding with temperature 0 and max 48 new tokens in the main run. Hidden states are extracted with `output_hidden_states=True`, using final-token residual vectors from depth-spanning selected layers (for QWEN2.5-1.5B-INSTRUCT: layers 1, 7, 14, 21, 28).

Environment: Python 3.12.8, PyTorch 2.10.0+cu128, Transformers 5.3.0, Datasets 4.4.0, NumPy 2.3.5, SciPy 1.17.1, scikit-learn 1.8.0, statsmodels 0.14.6. Hardware includes two RTX 3090 GPUs (24GB each) with bfloat16 inference.

3.4 Metrics

We evaluate:

- **Behavior:** exact-match accuracy for GSM8K numeric answers and COMMONSENSEQA option labels.
- **Drift:** cosine distance between draft and revised activations.
- **Geometry retention:** centered kernel alignment (CKA) between pre/post representation matrices.
- **Correctness separability:** logistic-probe AUC/F1 for correct vs. incorrect outcomes.
- **Trajectory stability:** round-to-round drift across three self-critique iterations.

3.5 Statistical Analysis and Baselines

The baseline is DIRECT prompting. We compare each revision condition to DIRECT with paired Wilcoxon signed-rank tests and apply Benjamini–Hochberg FDR correction across comparisons. Probe results are reported descriptively due limited sample size, especially for the small-model run where class imbalance makes AUC/F1 unstable.

4 Results

Main behavioral outcomes.

Table 1 shows that revision prompts do not consistently improve accuracy over DIRECT in this sample regime. On QWEN2.5-1.5B-INSTRUCT, EXTERNAL-CRITIQUE is best at 0.34 (vs. 0.26

Table 2: Paired significance tests on QWEN2.5-1.5B-INSTRUCT against DIRECT. No comparison is significant after FDR correction ($\alpha = 0.05$).

Comparison	Test	p	q (FDR)
REWRITE vs DIRECT	Wilcoxon	0.530	0.802
SELF-CRITIQUE-1 vs DIRECT	Wilcoxon	0.802	0.802
SELF-CRITIQUE-3 vs DIRECT	Wilcoxon	0.618	0.802
EXTERNAL-CRITIQUE vs DIRECT	Wilcoxon	0.359	0.802

Table 3: Deep-layer (layer 28) representation and separability summary on QWEN2.5-1.5B-INSTRUCT. AUC values show pre→post change from draft to revised condition.

Condition	Mean Cosine Drift	CKA (pre vs. post)	Probe AUC (pre → post)
REWRITE	0.3999	0.2354	0.818 → 0.545
SELF-CRITIQUE-1	0.3683	0.2034	0.818 → 0.750
SELF-CRITIQUE-3	0.4230	0.1890	0.818 → 0.889
EXTERNAL-CRITIQUE	0.3439	0.2445	0.818 → 0.300

for DIRECT), while SELF-CRITIQUE-3 drops to 0.22. On QWEN2.5-0.5B-INSTRUCT, absolute accuracy is lower overall, but the ranking is similar: EXTERNAL-CRITIQUE is highest at 0.25.

As shown in table 2, none of the QWEN2.5-1.5B-INSTRUCT pairwise comparisons is statistically significant after FDR correction (all $q = 0.802$). This means observed accuracy differences should be interpreted as directional, not conclusive.

Representation drift and geometry.

Table 3 indicates substantial deep-layer drift under all revision conditions (0.3439–0.4230 cosine distance). CKA values are low-to-moderate (0.1890–0.2445), suggesting meaningful geometric re-configuration. The largest drift occurs for SELF-CRITIQUE-3, while EXTERNAL-CRITIQUE preserves the highest pre/post geometric similarity.

Separability and convergence across rounds. Only SELF-CRITIQUE-3 improves deepest-layer correctness separability (AUC 0.818 → 0.889), whereas REWRITE and EXTERNAL-CRITIQUE reduce separability. This condition-specific result supports a partial, not universal, structured-drift claim.

Behavioral triangulation with API model. On GPT-4.1 ($n = 20$), DIRECT achieves 1.00 accuracy and SELF-CRITIQUE-1 reaches 0.95, with Wilcoxon $p = 1.0$. This small check further suggests that critique prompts do not guarantee behavioral gains even when base performance is high.

Ablation-style comparison by model size. Comparing QWEN2.5-1.5B-INSTRUCT and QWEN2.5-0.5B-INSTRUCT shows that the smaller model has much lower absolute accuracy and noisier probe behavior due class imbalance. The direction of condition ranking (EXTERNAL-CRITIQUE often strongest behaviorally) remains similar, but uncertainty increases markedly.

5 Discussion

What do the results imply? The key pattern is clear: internal-state movement is large, while behavioral gains are small and statistically inconclusive. This decoupling matters for evaluation. A prompt can induce substantial residual-stream drift without improving task accuracy in a reliable way.

When is drift “structured”? Our evidence is condition-dependent. SELF-CRITIQUE-3 shows both high drift and improved deep-layer separability, and its round-to-round drift decreases, which is consistent with convergence toward a more stable internal trajectory. However, this pattern does not hold for all revision prompts; EXTERNAL-CRITIQUE improves raw accuracy but weakens probe separability at the analyzed layer.

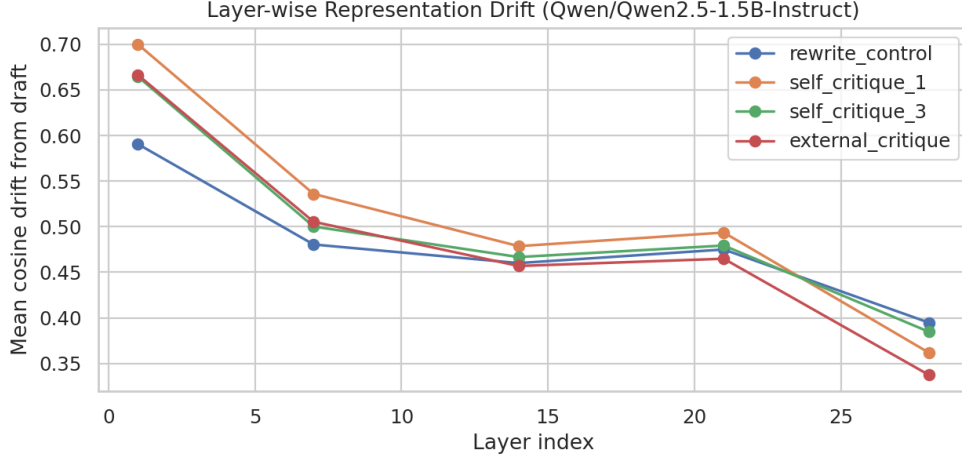


Figure 1: Layer-wise cosine drift on QWEN2.5-1.5B-INSTRUCT. Drift is non-trivial across depths and remains substantial in deeper layers.

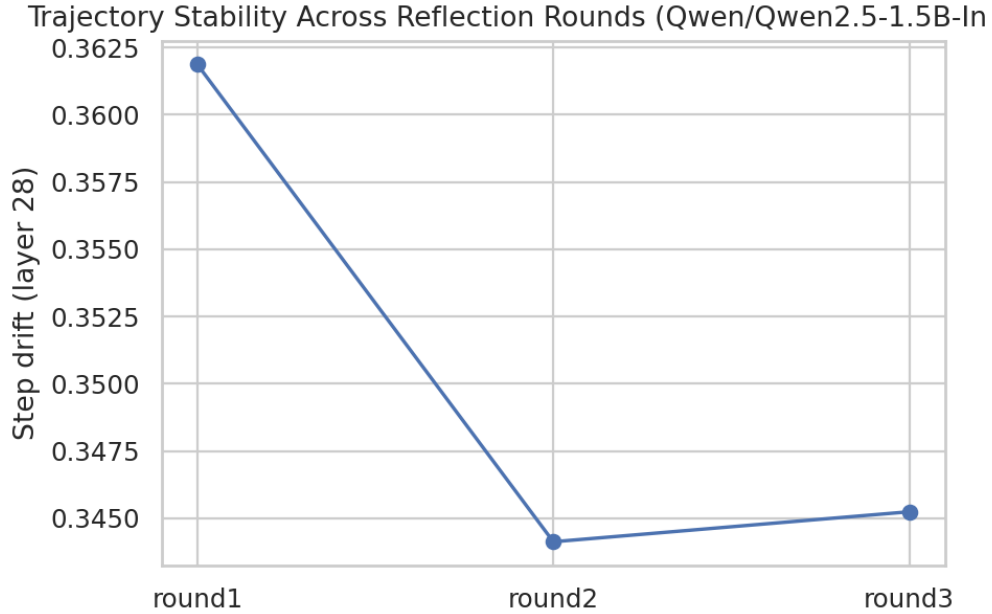


Figure 2: Self-critique trajectory on QWEN2.5-1.5B-INSTRUCT (layer 28). Step drift decreases across rounds (0.3683, 0.3347, 0.3359), consistent with partial convergence.

Limitations. Our main limitation is sample size (50 examples in the primary mechanistic run). We also analyze only final-token states, which can miss token-level trajectory structure. The study uses one run per condition set, so variance across random seeds and prompt perturbations remains unmeasured. The small-model probe estimates are unstable due to class imbalance. Finally, API experiments provide behavior only, not internals.

Broader implications. These findings suggest a practical protocol for reflection research: report mechanism and behavior separately, and test their coupling explicitly instead of assuming that one implies the other. For safety and reliability workflows, this separation can prevent over-claiming progress from prompt interventions that move representations but do not robustly improve correctness.

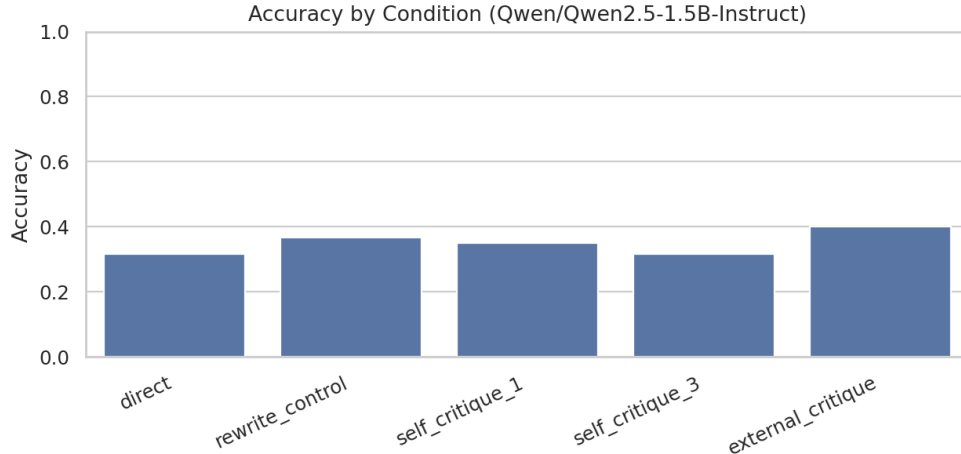


Figure 3: Accuracy by prompting condition for QWEN2.5-1.5B-INSTRUCT. Relative ordering changes across conditions, but significance tests are negative after correction.

6 Conclusion

We presented DRIFTCRITIQUE, a reproducible framework for evaluating self-critique through both output metrics and residual-stream dynamics. Across GSM8K and COMMONSENSEQA subsets, self-critique and revision prompts consistently produced substantial activation drift, while accuracy gains over DIRECT were not statistically significant in the primary run.

Our main takeaway is that representational change and behavioral improvement are distinct empirical axes. Multi-round self-critique showed signs of structured internal change (improved deep-layer separability and decreasing step drift), but this did not translate into reliable accuracy gains at current scale.

Future work should increase sample size (e.g., ≥ 200 per condition), add stronger controls (equal-token non-critique and shuffled-critique baselines), and analyze token-level trajectories with alignment tools such as SVCCA and Procrustes. Extending the same protocol to additional model families and truthfulness tasks will clarify when reflection-induced drift becomes predictively useful for performance.

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